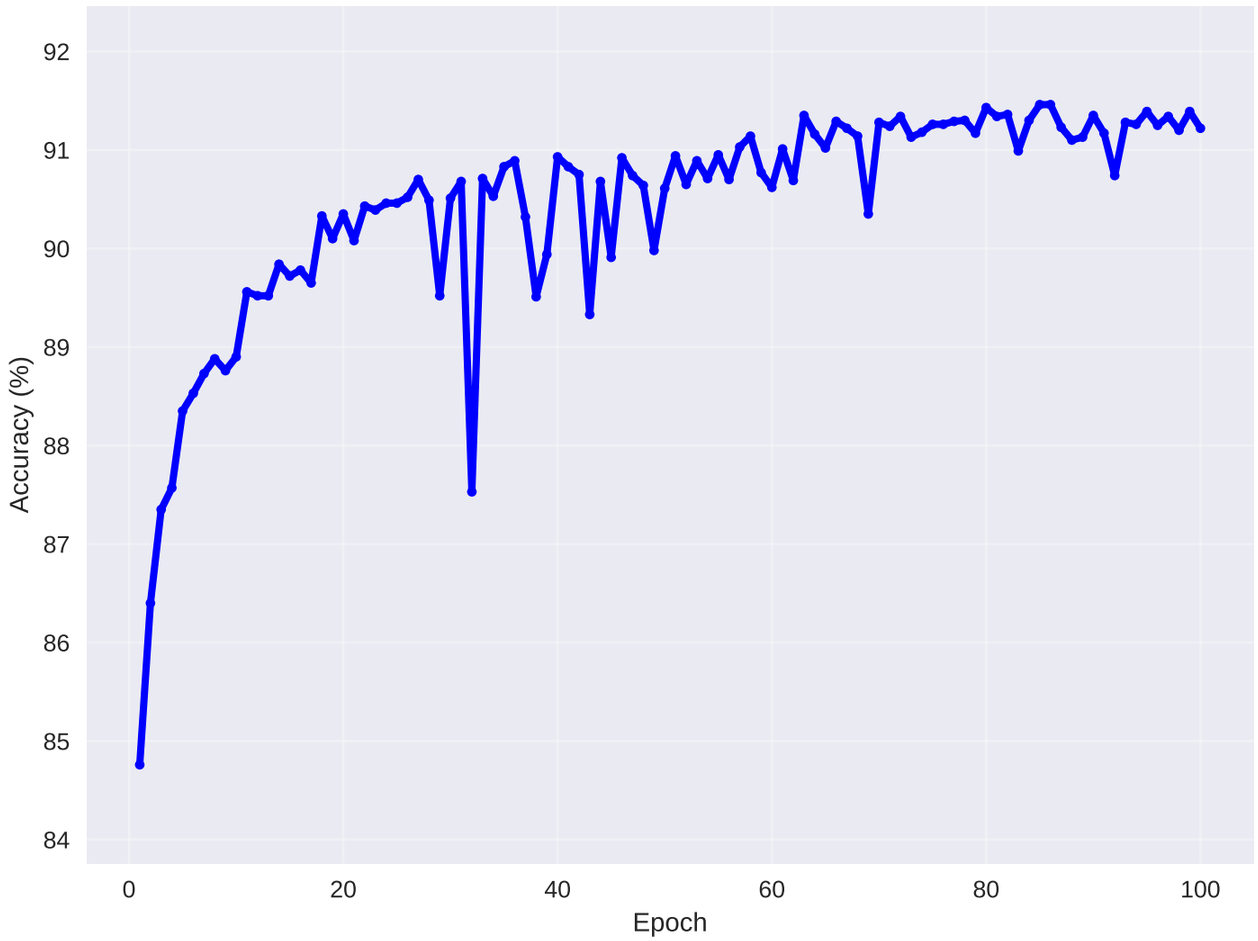
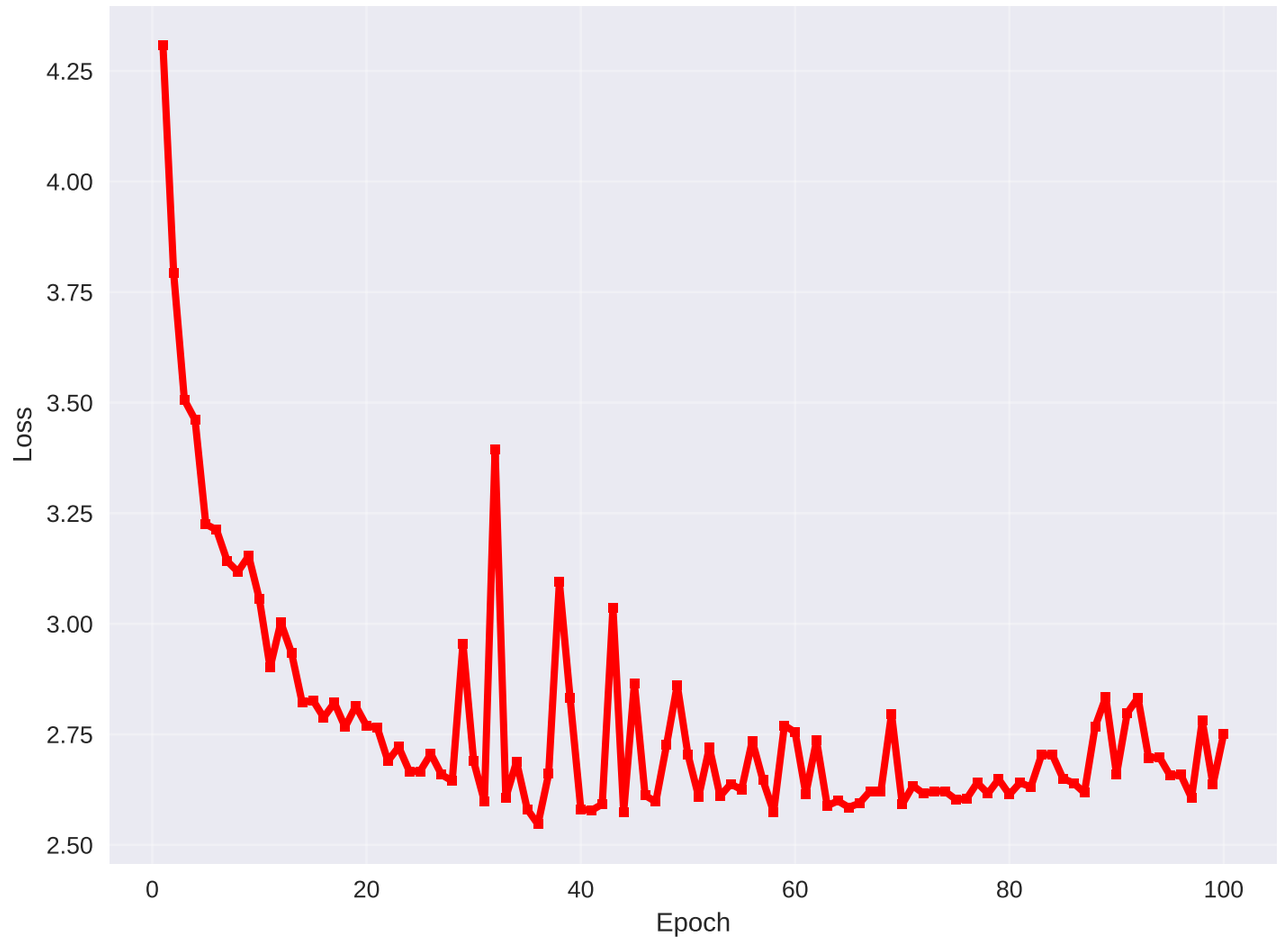
Federated Learning - Main Metrics Overview

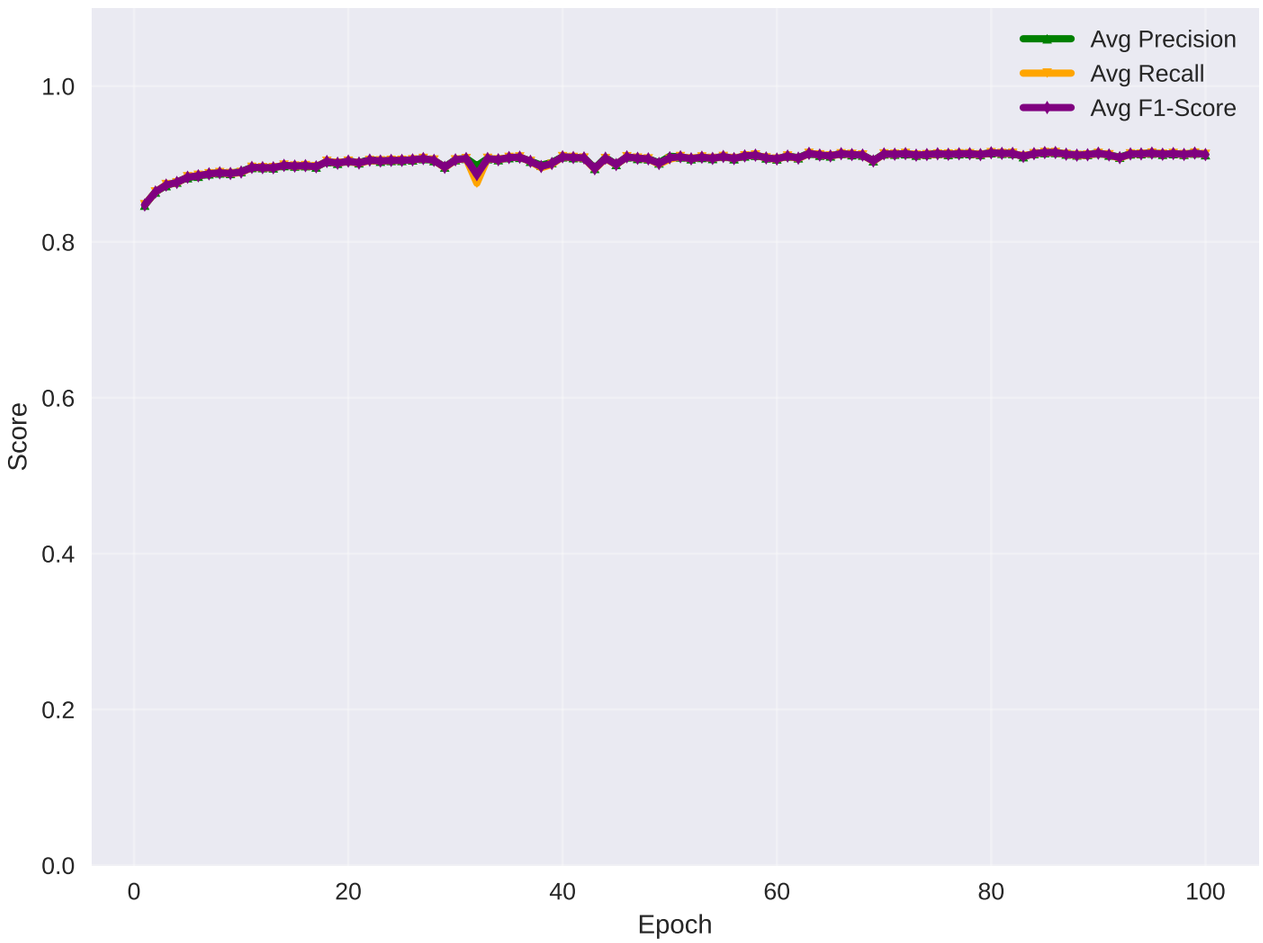
 Global Model Accuracy Evolution



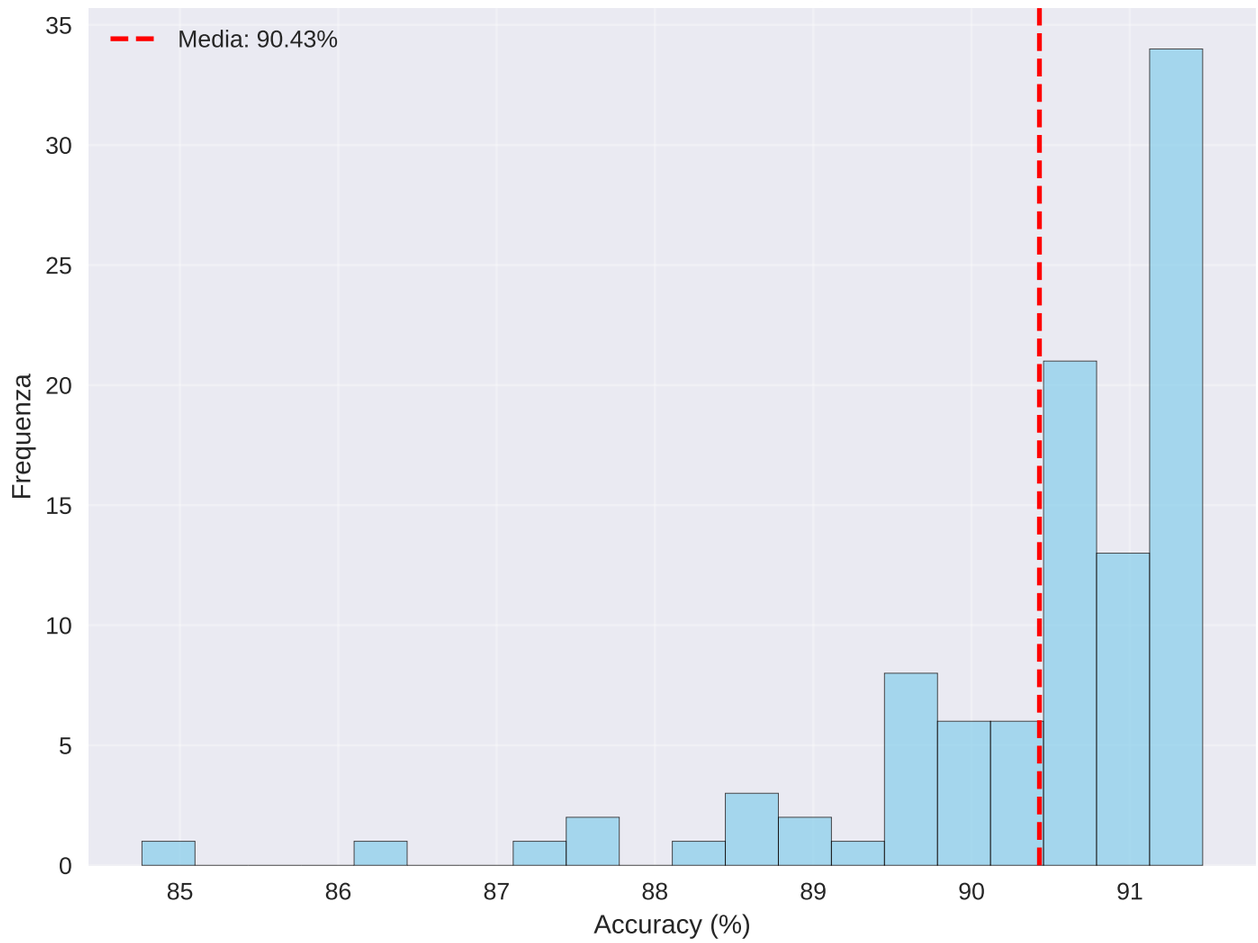
 Global Model Loss Evolution



 Average Precision, Recall & F1-Score

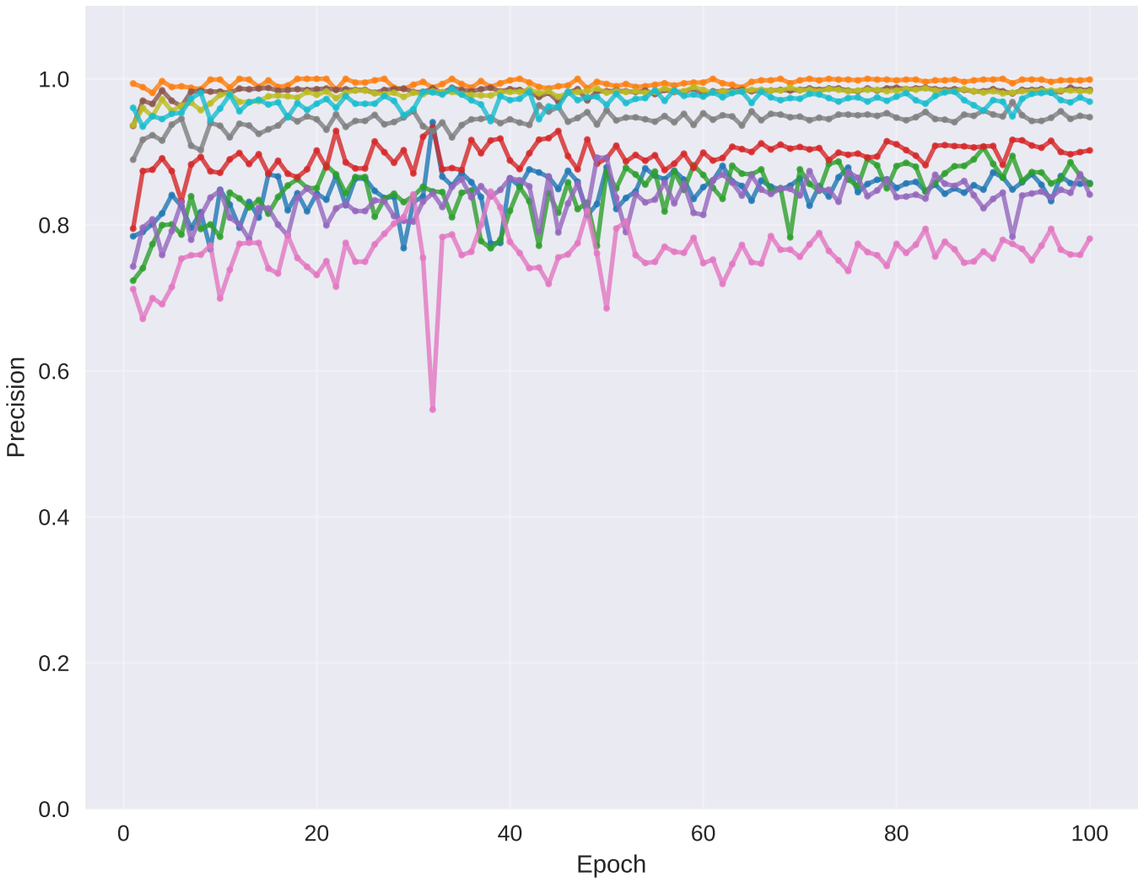


 Distribuzione Accuracy

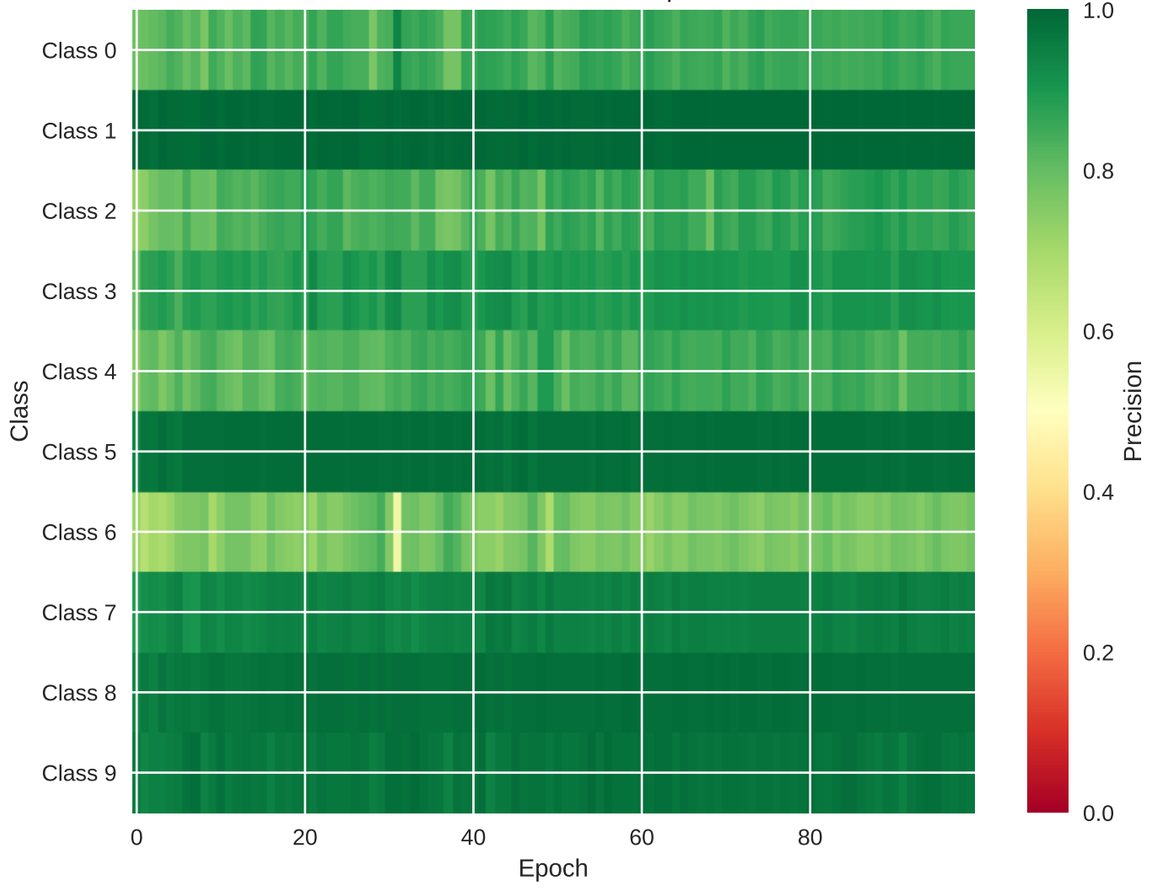


Per-Class Precision Analysis

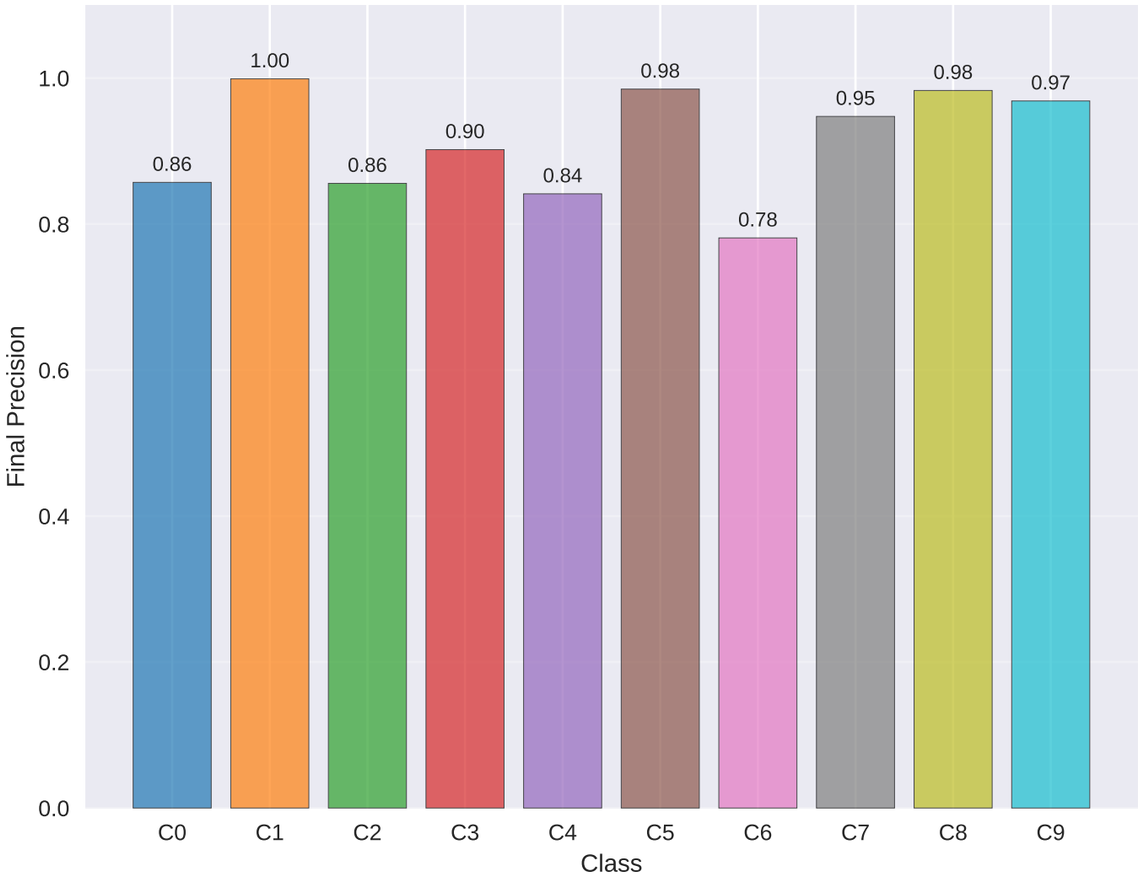
□ Per-Class Precision Evolution



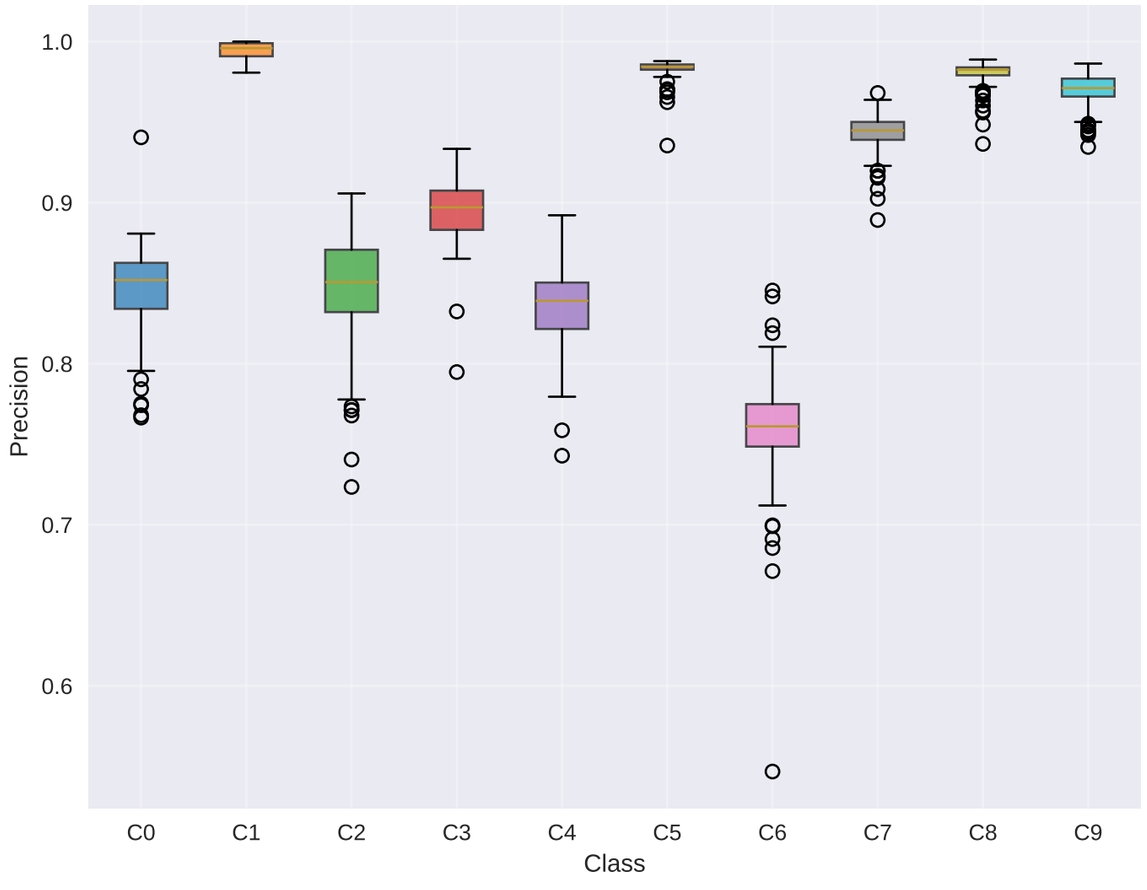
□ Precision Heatmap



□ Final Precision per Class

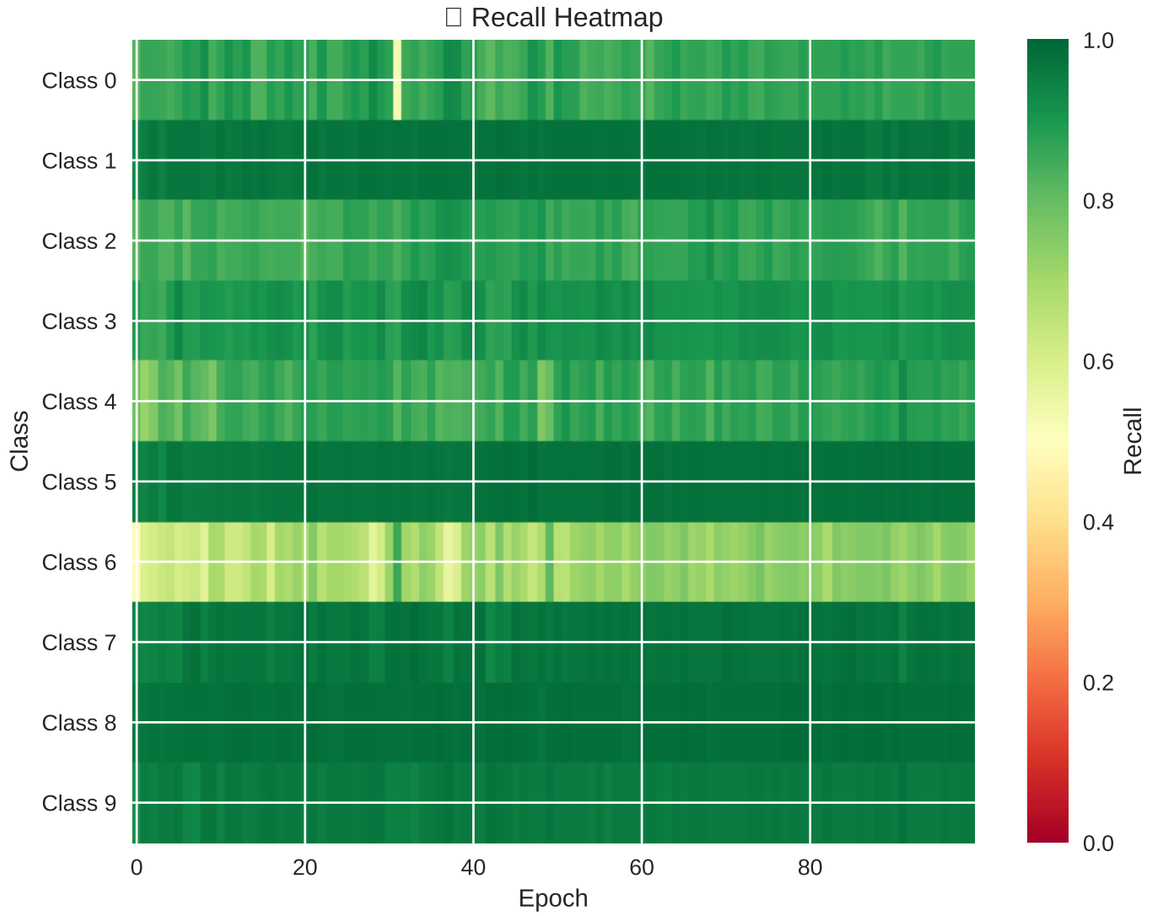


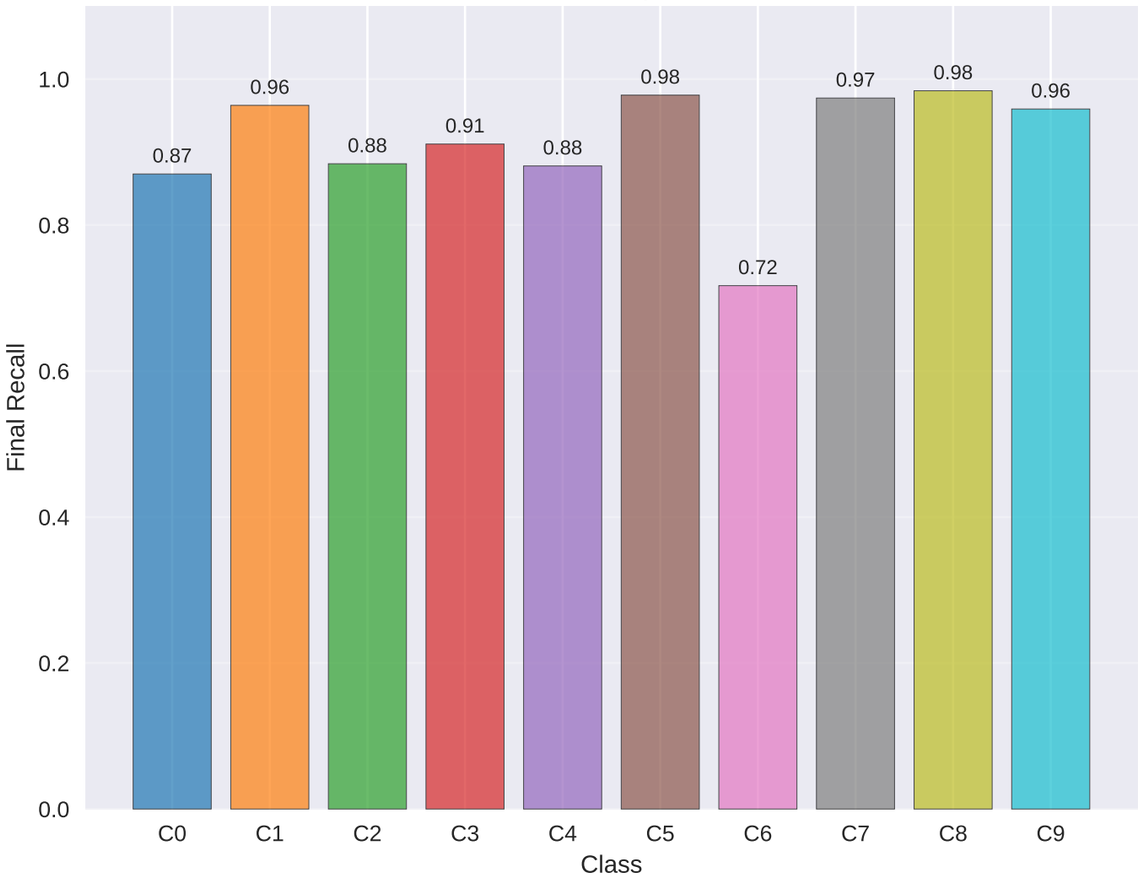
□ Precision Distribution per Class

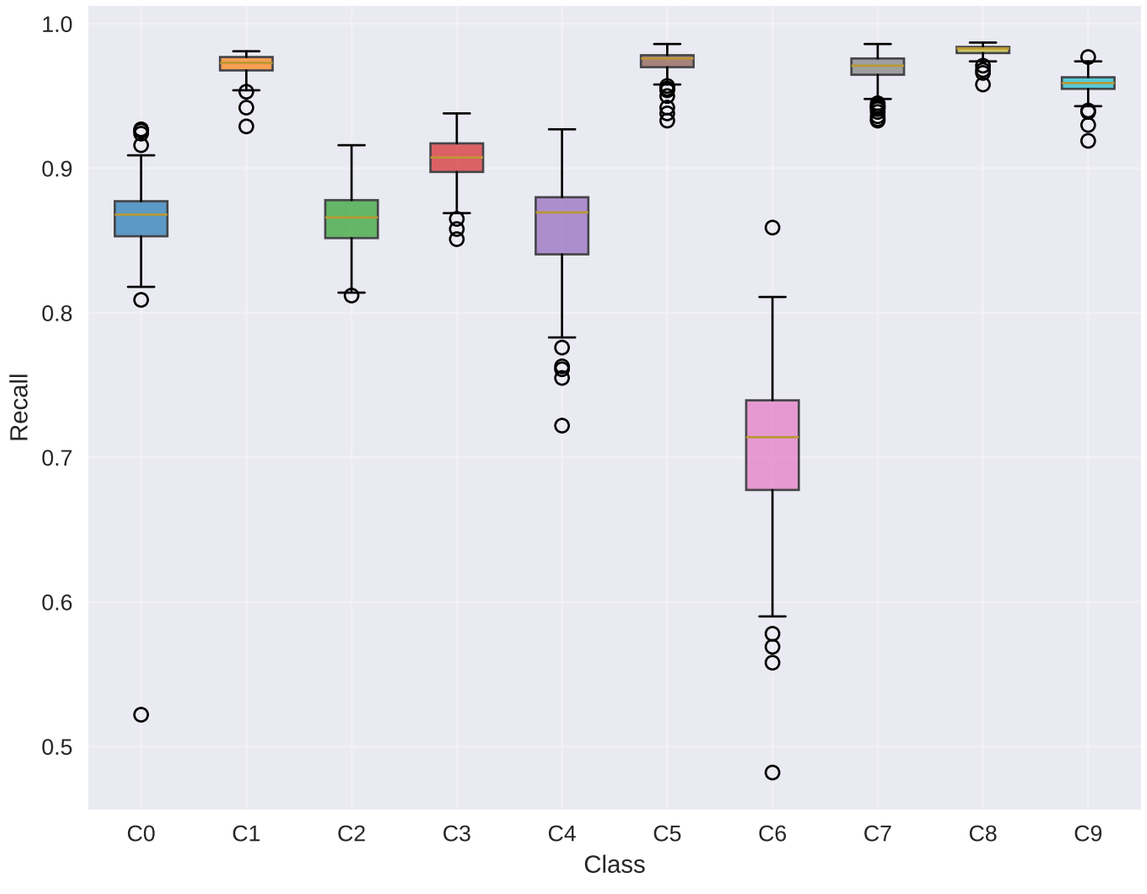


Per-Class Recall Analysis

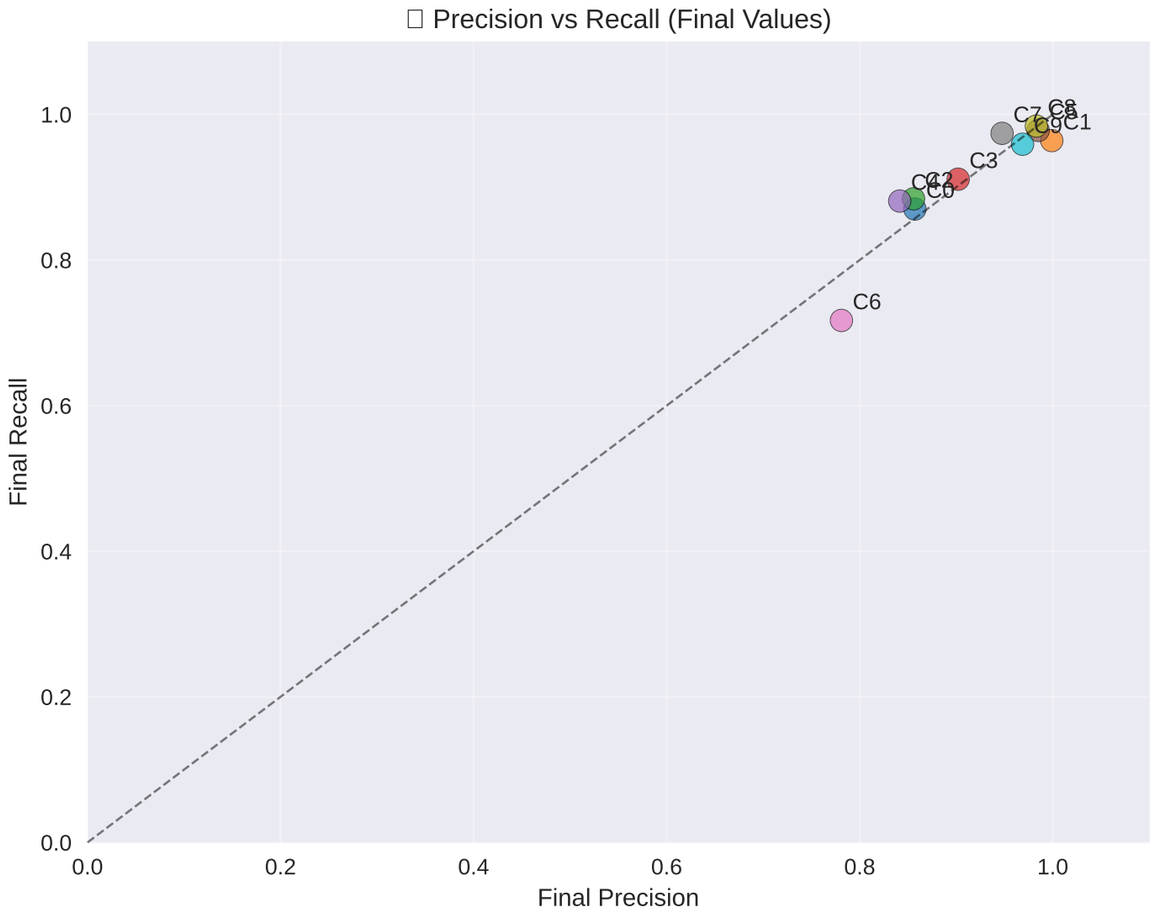






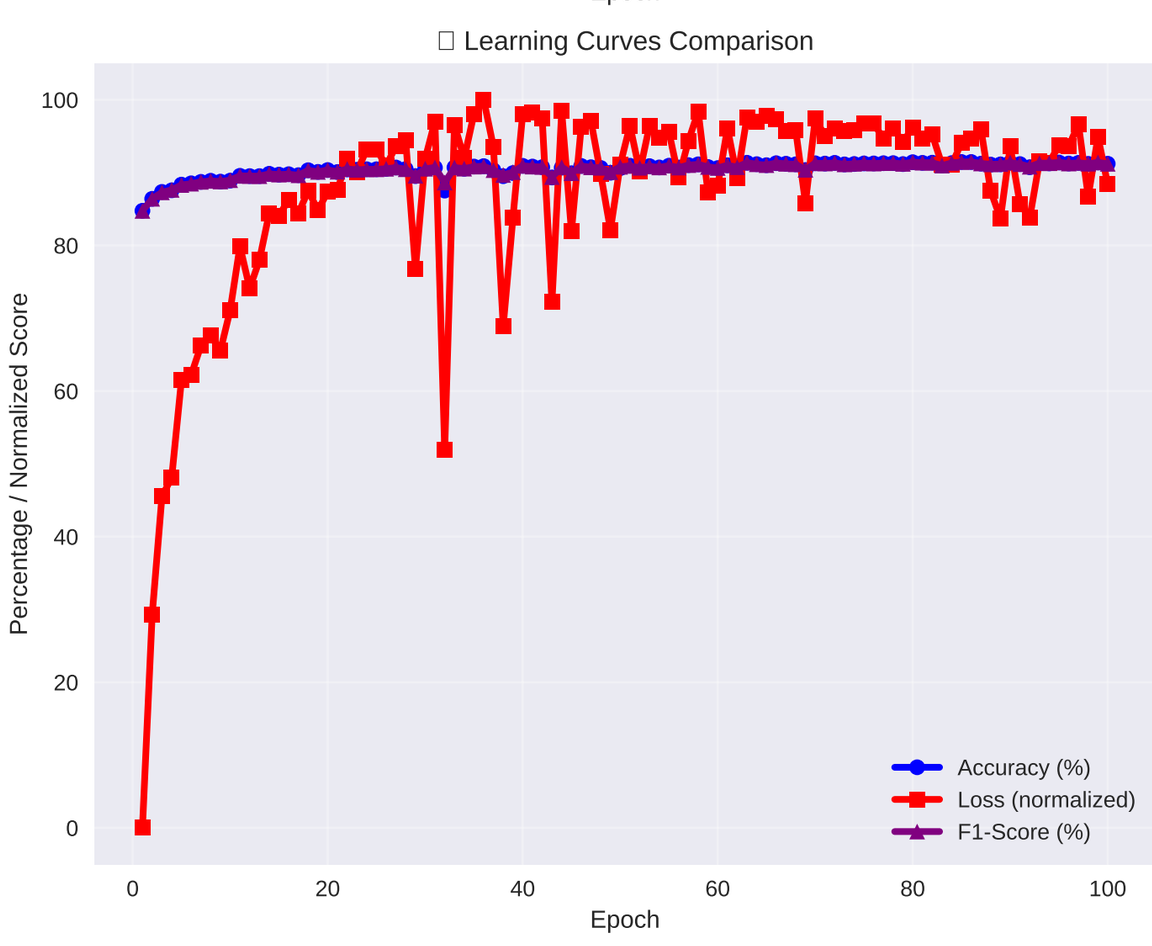


Comparative Analysis









Performance Summary Report

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FEDERATED LEARNING - PERFORMANCE SUMMARY
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FILE ANALIZZATO: 3000_results.csv
DATA ANALISI: 2025-08-30 01:43:53
EPOCHE TOTALI: 100

METRICHE FINALI:
• Accuracy: 91.22%
• Loss: 2.7509
• Average Precision: 0.912
• Average Recall: 0.912
• Average F1-Score: 0.912

EVOLUZIONE TRAINING:
• Miglioramento Accuracy: +6.46%
• Riduzione Loss: +1.5575
• Stabilità Accuracy (CV): 0.0127

PERFORMANCE PER CLASSE:
• Migliore Precision: Classe 1 (0.999)
• Peggiore Precision: Classe 6 (0.781)
• Migliore Recall: Classe 8 (0.984)
• Peggiore Recall: Classe 6 (0.717)

ANALISI DETTAGLIATA PER CLASSE:
• Classe 0: P=0.857, R=0.870, F1=0.864
• Classe 1: P=0.999, R=0.964, F1=0.981
• Classe 2: P=0.856, R=0.884, F1=0.870
• Classe 3: P=0.902, R=0.911, F1=0.906
• Classe 4: P=0.841, R=0.881, F1=0.861
• Classe 5: P=0.985, R=0.978, F1=0.981
• Classe 6: P=0.781, R=0.717, F1=0.748
• Classe 7: P=0.947, R=0.974, F1=0.961
• Classe 8: P=0.983, R=0.984, F1=0.984
• Classe 9: P=0.969, R=0.959, F1=0.964

VALUTAZIONE QUALITATIVA:
• Performance Globale: BUONA
• Stabilità Training: STABILE
• Balance Classes: SBILANCIATO

RACCOMANDAZIONI:
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